

Midterm Exam Solution

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Question 1

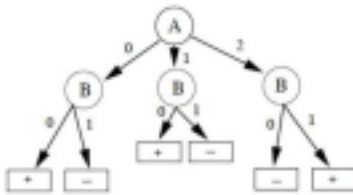


Figure 1: Decision tree for Question 1.

A	B	Number of + instances	Number of - instances
0	0	5	3
0	1	3	4
1	0	22	7
1	1	7	32
2	0	2	5
2	1	6	4

Table 1: Training set for Problem 1

A	B	Number of + instances	Number of - instances
0	0	4	1
0	1	3	1
1	0	6	3
1	1	3	15
2	0	5	2
2	1	2	5

Table 2: Test set for Problem 1

(a) [3 points] Gini index at node A
 $= 15/100 * \text{gini}(8,7) + 68/100 * \text{gini}(29,39) + 17/100 * \text{gini}(8,9) = 0.492$

(b) [3 points] Gini index at node b
 $= 29/68 * \text{gini}(22,7) + 39/68 * \text{gini}(7,32) = 0.325$

(c) [3 points] Estimate the generalization error rate of the tree using both the optimistic approach and the pessimistic approach. Use $\Omega = 2$ as the cost of adding a leaf node while calculating the pessimistic estimate.
 Error = $[(3 + 3 + 7 + 7 + 2 + 4) + 2 \times 6] / 100 = 0.38$

(d) [3 points] Compute the error rate of the tree on the test set shown in Table 4.
 Error test = $(1 + 3 + 3 + 3 + 5 + 5) / 50 = 0.4$

Question 1

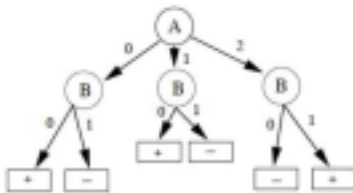


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Table 2: Test set for Problem 1

- (e) [3 points] Figure 2 shows a pruned version of the original decision tree. Estimate the generalization error rate of this tree using both the optimistic approach and the pessimistic approach, as in Part (a). Also compute the error rate of this tree on the test set shown in Table 4.

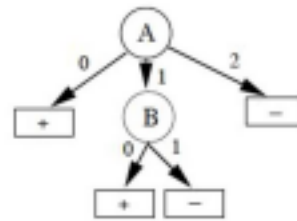


Figure 2: Pruned version of tree in Figure 1

Answer:

$$\text{Error}_{\text{generalization}} = [(7 + 7 + 7 + 8) + 2 \times 4] / 100 = 0.37$$

$$\text{Error}_{\text{test}} = (2 + 3 + 3 + 7) / 50 = 0.3$$

Question 2

(a) [2 points] 3 parameters

(b) [2 points]

1. What does an objective function $J(\theta)$ do ? A function to measure how “good” the linear model is
2. What are some properties of this function?
 - Should be differentiable
 - Preferably convex
3. What are some examples?
 - Mean Squared Error $\frac{1}{N} \sum_{i=1}^N e_i^2$
 - Mean Absolute Error: $\frac{1}{N} \sum_{i=1}^N |e_i|$

Question 2

(c) [3 points]

What is the gradient of the objective function , $J(\mathbf{w})$, w.r.t $\mathbf{w}$: $\nabla_{\mathbf{w}} J(\mathbf{w})$

$$\begin{aligned}\frac{dJ(\mathbf{w})}{dw_k} &= \frac{1}{5} \sum_{i=1}^5 -2x_k^{(i)}(y^{(i)} - \sum_{j=1}^3 w_j x_j^{(i)}) \\ \nabla_{\mathbf{w}} J(\mathbf{w}) &= \begin{pmatrix} \frac{dJ(\mathbf{w})}{dw_1} \\ \frac{dJ(\mathbf{w})}{dw_2} \\ \frac{dJ(\mathbf{w})}{dw_3} \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{5} \sum_{i=1}^5 -2x_1^{(i)}(y^{(i)} - \sum_{j=1}^3 w_j x_j^{(i)}) \\ \frac{1}{5} \sum_{i=1}^5 -2x_2^{(i)}(y^{(i)} - \sum_{j=1}^3 w_j x_j^{(i)}) \\ \frac{1}{5} \sum_{i=1}^5 -2x_3^{(i)}(y^{(i)} - \sum_{j=1}^3 w_j x_j^{(i)}) \end{pmatrix}\end{aligned}$$

Question 2

		Data points				
Attributes		$\mathbf{x}^{(1)}$	$\mathbf{x}^{(2)}$	$\mathbf{x}^{(3)}$	$\mathbf{x}^{(4)}$	$\mathbf{x}^{(5)}$
	x_1	1.0	2.0	3.0	4.0	5.0
	x_2	-2.0	-5.0	-6.0	-8.0	-11.0
	x_3	3.0	8.0	9.0	12.0	14.0
	y	2.0	4.0	7.0	8.0	11.0

(d) [5 points]

$$\frac{1}{5} \sum_{i=1}^5 -2x^{(i)}(y^{(i)} - \sum_{j=1}^3 w_j x_j^{(i)}) = \frac{-2}{5} \cdot (2 + 8 + 21 + 32 + 55) = -47.2$$

$$\frac{1}{5} \sum_{i=1}^5 -2x^{(i)}(y^{(i)} - \sum_{j=1}^3 w_j x_j^{(i)}) = \frac{-2}{5} \cdot (-4 - 20 - 42 - 64 - 121) = 100.4$$

$$\frac{1}{5} \sum_{i=1}^5 -2x^{(i)}(y^{(i)} - \sum_{j=1}^3 w_j x_j^{(i)}) = \frac{-2}{5} \cdot (6 + 32 + 63 + 96 + 154) = -140.4$$

$$\begin{aligned} \therefore \mathbf{w}^{(1)} &= \mathbf{w}^{(0)} - \alpha \nabla_{\mathbf{w}} J(\mathbf{w}) \\ &= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} - 0.1 \begin{pmatrix} -47.2 \\ 100.4 \\ -140.4 \end{pmatrix} \\ &= \begin{pmatrix} 4.72 \\ -10.04 \\ 14.04 \end{pmatrix} \end{aligned}$$

(e) [3 points] Yes, regularization

Question 3

Problem 3 (13 points)

(a) [3 points] overfitting

(b) [4 points] A, D

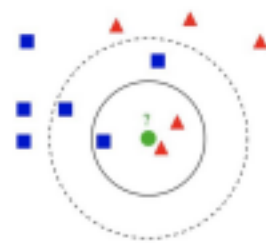
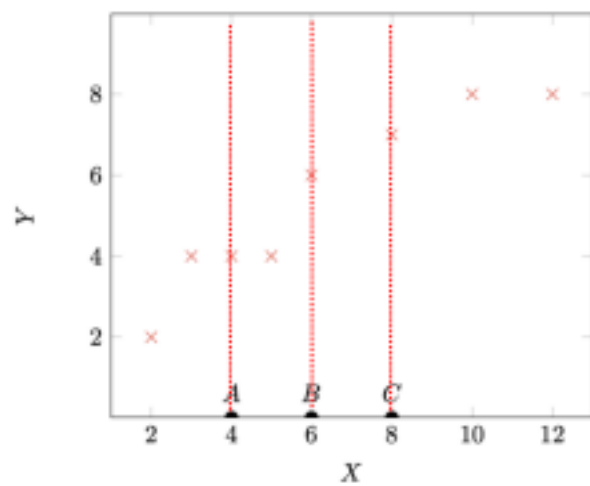
(c) [3 points] The model is overfitting. In order to address the problem, we can either increase training data size or decrease model complexity. We should never do (e), the model shouldn't see any testing data in the training process.

(d) [3 points] B. When model complexity increases, model can overfit better, so training error will decrease. But when it overfits too much, testing error will increase.

Question 4

- (a) • In the linearly separable case ($\vec{w} = \sum_i^n \alpha_i y_i \vec{X}^{(i)}$, $b = \sum_{i=1}^n \alpha_i y_i$, $\sum_{i=1}^n \alpha_i = 1$, $\alpha_i \geq 0$), all data points are on the correct side of the decision boundary and not within margin area; therefore if the removed point is on the margin ($\alpha_i > 0$), the decision boundary may shift toward the removed point after retraining, but if the removed point is not on the margin ($\alpha_i = 0$), the decision boundary will remain the same.
- (b)-(d) • For logistic regression, the decision boundary is decided by all data points; therefore, removing a point will eventually change the decision boundary.
- $\xi = 0$: the data point $\vec{x}_i$ and its label y_i satisfy $y_i(\vec{w}^T \vec{x}_i + b) \geq 1$, which means $\vec{x}_i$ is correctly classified.
 - $0 < \xi \leq 1$: the data point $\vec{x}_i$ and its label y_i satisfy $0 \leq y_i(\vec{w}^T \vec{x}_i + b) < 1$, which means $\vec{x}_i$ is correctly classified but within margin area.
 - $1 < \xi$: the data point $\vec{x}_i$ and its label y_i satisfy $y_i(\vec{w}^T \vec{x}_i + b) < 0$, which means $\vec{x}_i$ is not correctly classified (on the wrong side of the decision boundary).
- (e) The C controls the tolerance of error in the SVM model. If $C \rightarrow 0$, the SVM model will ignore any misclassification error. If $C \rightarrow \infty$, the SVM model will not tolerate any misclassification error.

Question 5



(a) $k=3$ triangle, $k=5$ square

(b) 0

(c) 4 and 5

(d) 7 and 5.375

Question 6

Problem 6 (12 points)

An instructor gives his class a true/false test and records the answers for each student as a binary vector. The instructor wants to see which students answered the questions in a similar way.

a) [4 points] Which of the following proximity measures would be appropriate for comparing student tests: the Jaccard measure, the simple matching coefficient (SMC) or the Hamming distance? Briefly explain.

Answer: Either the simple matching coefficient or the Hamming distance will work since they both count 1-1 and 0-0 matches. Hamming gives the number of mismatches, while SMC gives the percentage of matches as a fraction between 0 and 1.

b) [4 points] If the instructor gives a multiple choice test, how could you define a proximity measure to compare student tests?

Answer: Each question can be evaluated with respect to whether the answers match or not. Using this result, we can compute a similarity as computed by SMC or a distance as computed by the Hamming distance.

c) [4 points] Suppose the test consists of binary, categorical, and numerical answers. Would the measure you described for (b) still be appropriate? Briefly explain.

Answer: Yes, as long as you evaluate the similarity for numerical answers according to match / don't match.

Question 7

- (a) [5 points] Calculate the TPR and FPR of M for the + class for both these data sets. Is there a difference in their values for the two data sets?

$$\begin{aligned} \text{TPR}(\text{DS1}) &= 45/(45+5) = 0.9, \text{FPR}(\text{DS1}) = 10/(10+40)=0.2 \\ \text{TPR}(\text{DS2}) &= 45/(45+5) = 0.9, \text{FPR}(\text{DS2}) = 100/(100+400)=0.2 \\ \text{TPR and FPR are the same for M for the two data sets.} \end{aligned}$$

- (b) [5 points] Now, calculate Precision and Recall of M for the + class for both these data sets. Is there a difference in their values for the two data sets?

$$\begin{aligned} \text{P}(\text{DS1}) &= 45/(45+10)=0.818, \text{R}(\text{DS1})=45/(45+5)=0.9 \\ \text{P}(\text{DS2}) &= 45/(45+100)=0.31, \text{R}(\text{DS2})=45/(45+5)=0.9 \end{aligned}$$

Although the recall is same for the two data sets, precision is significantly different.

- (c) [5 points] What characteristic of the data sets (that are used to derive the above contingency tables) lead to the differences (if any) between the values of (TPR,FPR) and (Precision, Recall) that you observe above.

Answer: The class skew (or class imbalance) between the + and - classes is significantly different between the two data sets, i.e. 1:1 for data set 1 and 1:10 for data set 2. Since (TPR, FPR) are invariant to class skew, they don't change. However, Precision is not invariant to class skew, and thus its value changes.